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Molecular signature of human endometrial stem/progenitor cells at the single-cell level

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SUMMARY

Human endometrium sheds and regenerates each month during the menstrual cycle. N-cadherin$^+$ (*CDH2*) glandular epithelial progenitors and SUSD2$^+$ mesenchymal stem cells (MSCs) and their niches have been identified, but their signaling interactions remain unknown. SSEA1$^+$ epithelial cells resurface the endometrium, generating a new luminal epithelium each cycle. Using these markers, we characterized the gene expression of human endometrial stem/progenitor cell-enriched populations derived from fluorescence-activated cell sorting (FACS)-sorted hysterectomy endometrium by single-cell RNA sequencing (scRNA-seq). Two of 10 epithelial clusters contained *CDH2*$^+$*SOX9*$^+$ cells with high *TRH* and *IHH* expressions. N-cadherin and IHH, and SSEA1 and Hedgehog co-receptor BOC immunoco-localized in the basal layer of endometrial glands, from which the new functional layer glands regenerate each menstrual cycle. Two of six mesenchymal clusters contained *SUSD2*$^+$ MSCs, one with high *MUSTN1* expression. Epithelial progenitors and endometrial MSCs transitioned to their respective progeny. We provide new insights into the human endometrial stem/progenitor cell signaling pathways and niche interactions regulating their function.

INTRODUCTION

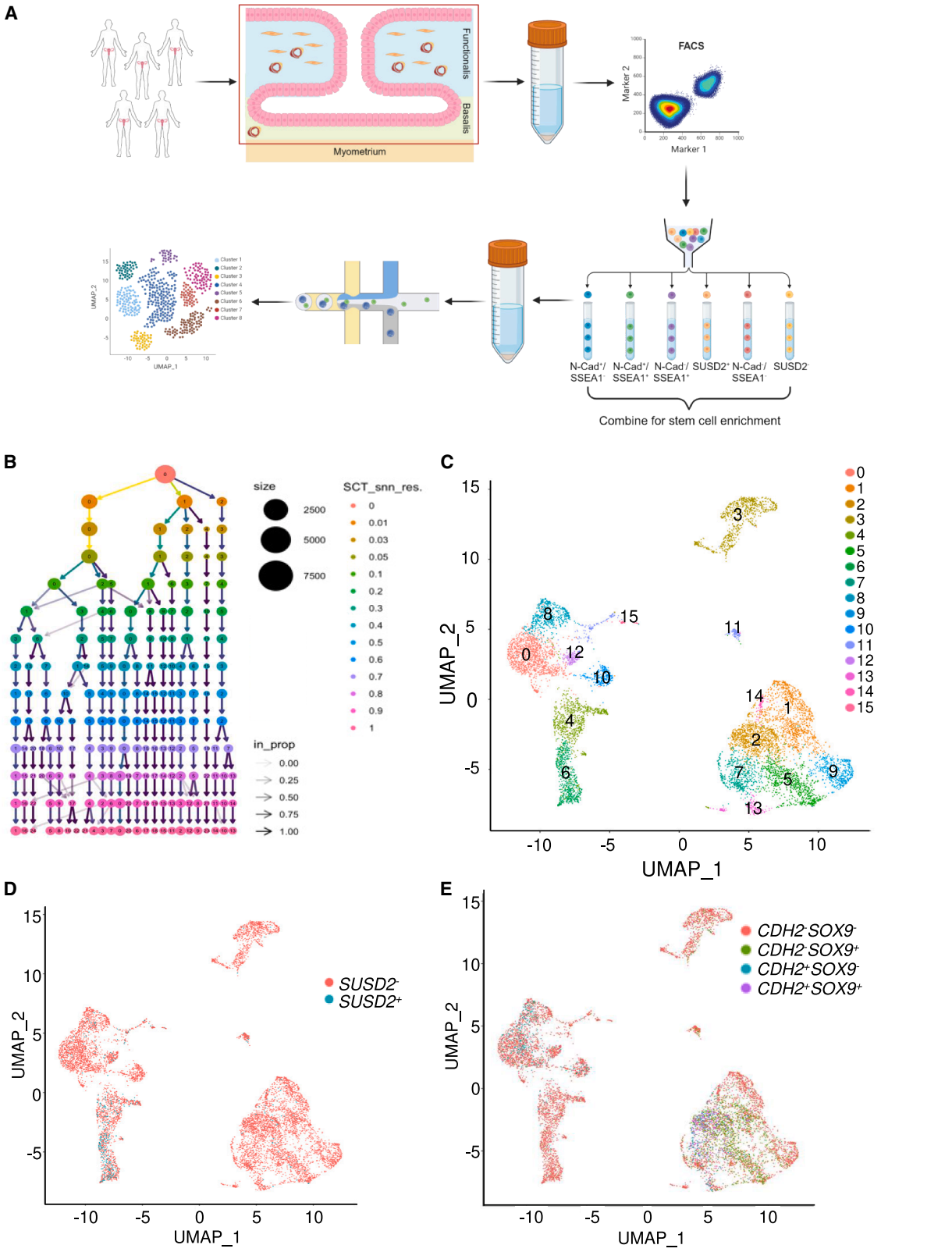
The endometrium (uterine lining) is the site where an embryo implants and the placentation occurs to support pregnancy. Human endometrium is one of the body's most highly regenerative tissues. During each menstrual cycle, the upper two-thirds (functionalis) is shed, leaving a raw surface on the remaining basalis, which rapidly re-epithelializes in the absence of the hormone estrogen (Gargett et al., 2012; Kaitu'u-Lino et al., 2010). As circulating estrogen levels rise during the proliferative stage of the cycle, a new functionalis regenerates, with glands sprouting vertically from the horizontal, interconnected glands of the basalis (Yamaguchi et al., 2022). Following ovulation, during the secretory phase, ovarian progesterone induces gland differentiation and histotroph secretion to nourish an embryo during early placentation (Burton et al., 2002). In the absence of an implanting embryo, circulating progesterone falls, and the functionalis is shed piecemeal into menstrual fluid. Up to 1 cm of the endometrial tissue regenerates each month, comprising glands lined by columnar epithelium and a vascularized stroma. More than 400 menstrual cycles occur in modern reproductive-age women (Salamonsen et al., 2021).

The first evidence for human endometrial stem/progenitor cells was the demonstration of clonogenic epithelial and stromal cells derived from hysterectomy endometrium (Chan et al., 2004). Single cells initiating large epithelial or stromal clones undergo self-renewal (serial cloning), generating $\sim 6 \times 10^{11}$ cells. Epithelial clones differentiate into 3D gland-like structures, and large stromal clones differentiate into multiple mesodermal lineages *in vitro* (Gargett et al., 2009), indicating that the human endometrium contains epithelial progenitors and mesenchymal stem cells (MSCs).

The first markers enriching for clonogenic endometrial MSCs (eMSCs), co-expression of CD140b and CD146 (Schwab and Gargett, 2007), identified their perivascular niche in both the functionalis and basalis. A single perivascular marker, SUSD2, also enriched for clonogenic MSCs and differentiated into stromal tissue and endothelium *in vivo* (Masuda et al., 2012).

CDH2 encoding N-cadherin was first identified as a human endometrial epithelial progenitor marker by its upregulation in post- menopausal relative to pre-menopausal hysterectomy-derived endometrial epithelial cells (Nguyen et al., 2017). Purified N-cadherin$^+$ epithelial cells showed greater epithelial progenitor activity than N-cadherin$^-$ epithelial cells and localized to the horizontal branching glands next to the myometrium of the pre-menopausal basalis. Post-menopausal endometrium can regenerate a functionalis on exogenous estrogen

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exposure. AXIN-2 and SSEA1 were discovered as basalis epithelial markers (Nguyen et al., 2012; Valentijn et al., 2013). SSEA1$^+$ basalis epithelial cells were identified as having intermediate telomerase activity and longer telomeres, suggesting progenitor activity. SSEA1$^+$ epithelial cells are nuclear SOX9$^+$ and located at an ill-defined basalis-functionalis junction, where they migrate from basalis gland stumps during menses to resurface the raw endometrium and become the new luminal epithelium (LE) (Cousins et al., 2021).

Recent single-cell RNA sequencing (scRNA-seq) studies have identified multiple endometrial epithelial, stromal, and immune cell subpopulations in human endometrium (Fonseca et al., 2023; Garcia-Alonso et al., 2021; Lv et al., 2022; Marečková et al., 2024; Queckbörner et al., 2021; Tan et al., 2022; Wang et al., 2020). Some studies have identified perivascular eMSCs present in the functionalis of the biopsies sampled and found our three published eMSC markers (Queckbörner et al., 2021; Tan et al., 2022; Wang et al., 2020). *CDH2*$^+$ epithelial progenitors were mostly not detected due to their deep basalis location not sampled in a biopsy, although *AXIN2*$^+$ cells were found. Two clusters comprising many *SOX9*$^+$ epithelial cells—40-fold more than *SOX9*$^+$ basalis and luminal epithelial (LE) cells—were found in the functionalis glandular epithelium during the proliferative phase (Marečková et al., 2024). *SOX9*$^+$ cells appear more numerous than expected for a basalis and luminal epithelial marker (Garcia-Alonso et al., 2021).

To gain insight into the stem/progenitor cell state of epithelial progenitors and eMSCs and their respective niche cells that regulate cell fate decisions, we enriched human endometrial stem/progenitor cells by fluorescence-activated cell sorting (FACS) of hysterectomy-derived endometrial single-cell suspensions, using our surface markers, N-cadherin, SSEA1, and SUSD2, and recombined them with their mature progeny for subsequent scRNA-seq.

RESULTS

Cell identities defined by scRNA-seq in human endometrial stem/progenitor cell-enriched populations

To generate a molecular signature of human endometrial stem/progenitor cells, we first isolated cells from hysterectomy endometrium containing basalis and functionalis from 5 women not taking hormones. Single-cell suspensions were enriched for epithelial progenitor cells and eMSCs by FACS, using known markers (Figure S1), generating 6 subpopulations of endometrial cells: 3 basalis EpCAM$^+$ epithelial progenitors, N-cadherin$^+$SSEA1$^-$, N-cadherin$^+$SSEA1$^+$, and N-cadherin$^-$SSEA-1$^+$; mature epithelial cells, EpCAM$^+$N-cadherin$^-$SSEA1$^-$; eMSCs, SUSD2$^+$ EpCAM$^-$ and stromal fibroblasts, SUSD2$^-$EpCAM$^-$. The 6 subpopulations were pooled to enrich for stem/progenitor cells and subjected to scRNA-seq (Figure 1A). An average of 2,346 cells were sequenced per sample, with a median of 3,105 genes detected per cell (Table 1). Following sample integration, cell clustering was performed at 0.4 resolution based on cluster stability and identification of known biological groups and cell types (Figure 1B). Sixteen cell clusters were identified (Figure 1C). Cluster relationships varied with resolution, reflecting differences in cell type, state, and lineage commitment. Cell numbers contributed by each subject within a cluster varied (Figure S2A), likely due to the cycle stage at tissue harvest, individual pathologies (Table S1), and individual variation in the cellular composition. However, all samples contributed cells to each cluster (C) and clusters aligned with the original pooled populations (Table S2).

Each cell was annotated to a cell type based on similarity in expression profiles with a reference dataset (Aran et al., 2019) (Figure S2B). Clusters were assigned a cell type based on the expression of previously reported markers for endometrial cell types (Figure 2, C0–C16). Ten clusters were of epithelial origin (clusters 2, 7, 13, 1, 5, 9, 14, 3,

Figure 1. scRNA-seq analysis of human endometrial stem/progenitor cell-enriched endometrial cells

(A) Workflow from the collection of ($n = 5$) endometrial tissues from subjects undergoing hysterectomy, isolation of cells from the basalis and functionalis glands and stroma (red box, without myometrium), and FACS of published stem/progenitor cell markers, recombined in equal proportions where possible and analyzed by scRNA-seq.

(B) Clustree visualization of clusters generated by Seurat's Louvain algorithm for varying resolution parameters (rows). Each circle represents a cluster identified at a given resolution, with size corresponding to the number of cells contained. Arrows between clusters illustrate cell transitions across resolutions, with arrow shading showing the proportion of cells carried over from clusters at the preceding resolution.

(C) UMAP showing 16 defined cell clusters at 0.4 resolution.

(D) UMAP of *SUSD2*$^+$ eMSC and *SUSD2*$^-$ mesenchymal cells.

(E) UMAP of *CDH2* and *SOX9* expression in the cell clusters. Clusters 2 and 7 contain higher numbers of *CDH2*$^+$ and *SOX9*$^+$ cells, likely endometrial epithelial progenitor cells.

UMAP, uniform manifold approximation and projection.

See also Figures S1–S3 and Tables S1–S4.

Table 1. Single-cell RNA sequencing metrics for each sample

Donor ID	Estimated number of cells	Mean reads per cell	Median genes per cell	Reads mapped to genome (%)	Total genes detected	Median UMI counts per cell
1	1,460	623,754	3,529	96.0	26,874	13,890
2	3,797	309,348	2,892	96.7	27,044	9,946
3	2,520	656,319	2,243	95.6	26,402	7,067
4	3,336	171,518	3,147	95.8	27,066	10,216
5	618	162,440	3,714	96.0	22,387	16,654

15, and 11) based on *EPCAM* expression (Figure S3A), and six were of mesenchymal origin (clusters 4, 6, 12, 0, 10, and 8).

Human endometrial stroma contains a subpopulation of perivascular cells fulfilling designated MSC properties as per the International Society for Cell and Gene Therapy (Viswanathan et al., 2019) and by their stem cell functional properties (Masuda et al., 2012). Here, we identified eMSCs based on well-characterized perivascular markers SUSD2 (Masuda et al., 2012; Sivasubramaniyan et al., 2013), MCAM (Crisan et al., 2008; Schwab and Gargett, 2007), and RGS5 (Bondjers et al., 2003). A total of 520 *SUSD2*$^+$ cells were identified (Table S3). *SUSD2* (Figures 1D and 2), *MCAM*, and *RGS5* (Figure 3A) expression was selectively enriched in the cells in C6 and C4. A presumptive marker for endometrial perivascular cells from prior endometrial scRNA-seq studies, *STEAP4* (Garcia-Alonso et al., 2021; Marečková et al., 2024), was confirmed in the C4 eMSC-1 but not C6 eMSC-2 subpopulations, but it was also present in the epithelial populations C1 and C14 (Figures 2 and S3C). A marker for myometrial perivascular cells, *MYH11* (Garcia-Alonso et al., 2021; Marečková et al., 2024), showed high expression in C6 eMSC-1 but also in C10 decidualized stromal-2 cells (Figures 2 and 3A). eMSCs were observed in all samples (Figures S3G and S3H; Tables S3 and S4).

Typical mesenchymal genes, *PDGFRB* (Schwab and Gargett, 2007) and *ACTA2*, were identified in all 6 mesenchymal clusters at varying expression levels (Figure 2). In contrast, *PDGFRA* and *IGF1* expressions were selectively enriched in the non-eMSC clusters 12, 0, 10, and 8. We confirmed *C7* as a basal fibroblast gene (C12) (Figure 2) (Garcia-Alonso et al., 2021; Marečková et al., 2024). Decidualization, essential for embryo implantation, is a terminal differentiation of functionalis endometrial stromal cells into an epithelioid phenotype during the secretory stage (Gellersen and Brosens, 2014). Two clusters of stromal cells expressing decidualization genes were identified, C0 and C10. Defining genes for C0 were *IL15* and the WNT inhibitor *DKK1*, and that for C10 was *LEFTY2* (Figure 2). LEFTY2 is likely a negative regulator of the window of implantation and decidualization. Its high expression in C10 and C8 indicates stromal cells in a late secretory phase (Salker et al., 2018). Late secretory-stage stroma (C8) is defined by a high expression of *CXCL8*, which is also highly expressed in C9 secretory glandular and C5 junctional epithelial cells (Figure 2). The proportion of donor cells assigned to secretory (C1, C5, C9, and C8) and decidualized stroma (C0 and C10) cell type clusters was consistent with two donors in the secretory phase and one in the periovulatory phase (Table S5; Figure S3G). The percentage of cells positive for specific cell type markers contributed by each donor mostly also reflected their individual cycle phase (Table S6).

Epithelial progenitors were identified on the basis of their expression of published specific protein markers that enriched for cells with functional stem/progenitor cell properties (Nguyen et al., 2017; Salamonsen et al., 2021). A total of 796 *EPCAM*$^+$*CDH2*$^+$ epithelial progenitors were identified in 2 clusters (Table S3). Of these, 476 cells were *EPCAM*$^+$*CDH2*$^+$*SOX9*$^+$ (Table S4). *CDH2*, *SOX9*, and *AXIN2* were concentrated in C2 and extended into C7 (Figures 1E, 2, S3B, and S3E). Clonogenic, self-renewing EPCAM$^+$N-cadherin$^+$SOX9$^-$ cells were previously found in the basalis glands next to the myometrium (Nguyen et al., 2017). By scRNA-seq, rare populations of *CDH2*$^+$*SOX9*$^-$ cells were found to be scattered mainly in C2 and C7 (Figure 1E) and in several stromal clusters, which may be *EPCAM*$^-$*CDH2*$^+$ stromal cells (Figure 1E). *EPCAM*$^+$*CDH2*$^+$*SOX9*$^+$ cells were almost exclusively localized to C2 and C7 (Figure 1E), suggesting they are basalis glandular epithelial progenitor cells, which may translate *CDH2* to N-cadherin but not SOX9 proteins. They were mainly found in the proliferative phase samples from donors 1 and 2 (Figure S3I). *EPCAM*$^+$*CDH2*$^-$*SOX9*$^+$ cells, mostly present in C2 and C5, are likely luminal or glandular progenitor cells. Mature epithelial *EPCAM*$^+$ *CDH2*$^-$*SOX9*$^-$ cells were present within most epithelial clusters except C2.

EPCAM$^+$ cells in C2, C7, C13, C1, and C5 (Figures 2 and S3D) are likely glandular epithelial cells due to their higher *FOXA2* expression (Kelleher et al., 2019). C5 cells with high

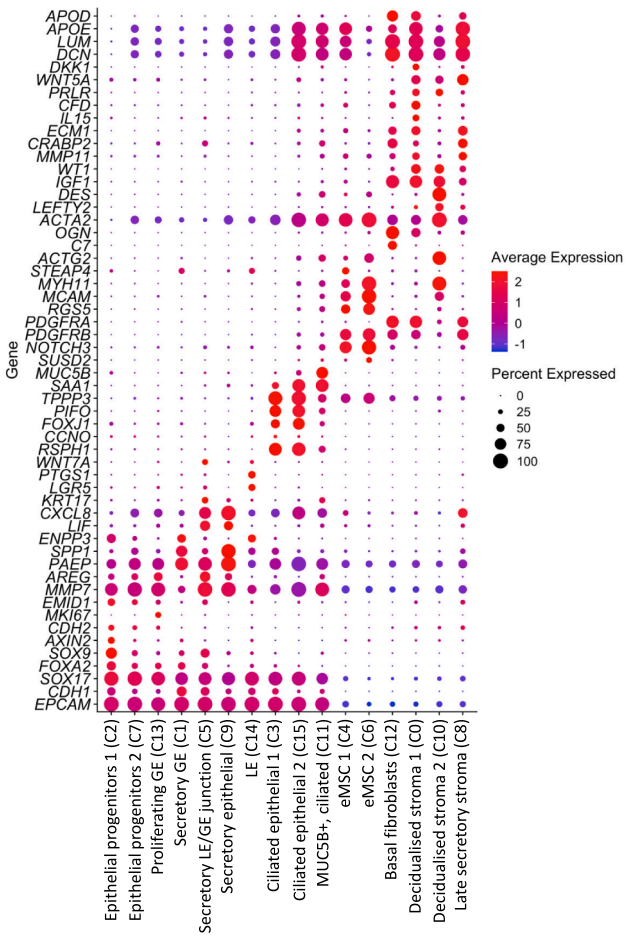


Figure 2. scRNA-seq identified 16 endometrial cell populations
Dot plot showing previously published markers of endometrial cell types used to annotate the cell populations.
See also Figure S3 and Tables S2–S6.

FOXA2 expression also expressed the secretory phase markers *LIF*, *PAEP*, and *SPP1* and the LE markers *SOX9*, *KRT17*, *WNT7A*, and *LGR5* (Garcia-Alonso et al., 2021), suggesting junctional zone cells between the glandular epithelium (GE) and the LE (Figure 2). C1, also with high *FOXA2* expression and progesterone-responsive markers *PAEP, SPP1*, and *ENPP3*, suggested secretory GE cells (Figure 2). C9 and C14 had lower *FOXA2* expression. C9 also expressed secretory markers, suggesting another secretory LE. C14 cells expressing *LGR5* and *ENPP3* may be a mix of secretory LE and LE cells (Figure 2). High *MKI67* expression in C13 indicated proliferating GE cells. Epithelial C3, C15, and C11 with high proportions of cells expressing *EPCAM, FOXJ1*, and *PIFO* were annotated to ciliated epithelial cells (Figure 2). C11 showed a high expression of *MUC5B*, once reported to define epithelial progenitors (Tan et al., 2022) but more likely are endocervical cells (Marečková et al., 2024).

eMSC populations identified in human endometrium

Highly enriched genes within eMSC clusters C4 and C6 included the classic MSC/eMSC signaling markers *NOTCH3* and *RGS5*, the transcription factor *EBF1*, and *CPE*, which cleaves C-terminal basic residues of prohormones (Figure 3A). The *GUCY1A2* subunit involved in generating cGMP expressed in 75% of C4 cells and 11.6% of cells in all other clusters, highlighting specificity for C4. MSC markers *MYH11* and *MCAM* were found in C6 cells. *MUSTN1*, a development-dependent smooth and skeletal muscle regulatory gene involved in growth (Hadjiargyrou, 2018), showed the highest specificity of expression (93.9% vs. 6.6%) and greatest fold change (3.16) in C6 cells (Figure 3A; Data S1). *NOTCH3* was most highly and specifically expressed in C6 *SUSD2*$^{+}$ and to a lesser extent in C4 cells (Figures 3A and 3B, Data S1). The ligand *JAG1*, bHLH family transcription factor transcripts *HES4, HEYL*, and *HEY2*, the transcriptional co-activator *MAML2*, and the γ-secretase subunit *APH1A* were also expressed, indicating active canonical NOTCH signaling (Weber et al., 2014) that regulates cell fate decisions between eMSCs and endometrial niche cells. Biological processes enriched for genes in C4 eMSCs included wound healing and vasculature development (Figure S4A). Genes expressed in C6 were also enriched for vasculature development, actin cytoskeletal organization, and muscle development. In contrast, more mature decidualized stroma C0 and late secretory stroma C8 were enriched for processes related to differentiated cells such as cellular response to amino acid stimulus and extracellular matrix organization (Figure S4B).

RNA velocity indicates that C6 and C4 cells are stem/progenitor cells (Figure 3C), particularly the *SUSD2*$^{+}$ eMSCs (Figure S5A). Using the predicted C4 and C6 progenitor cells as root populations in cell fate trajectory analysis showed that this cell state differentiated toward a more mature fibroblast state in C0, C8, and C12 (Figure 3C). Cell cycle phase analysis revealed that most cells in each mesenchymal cell cluster were in G1, with the least in C4 eMSC-1 followed by C6 eMSC-2, suggesting that eMSCs are more proliferative than fibroblasts (Figure S5D).

Endometrial epithelial progenitor populations identified in human endometrium

Key markers of epithelial progenitors, GE and LE were examined in the epithelial clusters (Figure 4A). Highly enriched genes for the two epithelial progenitors, C2 and C7, included the transcription factors *SOX9* and *SOX17* (Figures 2 and 4A) and the developmental genes *PRR15* and *CPM*, which cleave basic C-terminal residues from peptides/proteins with potential roles in controlling peptide hormone and growth factor activity at the cell surface

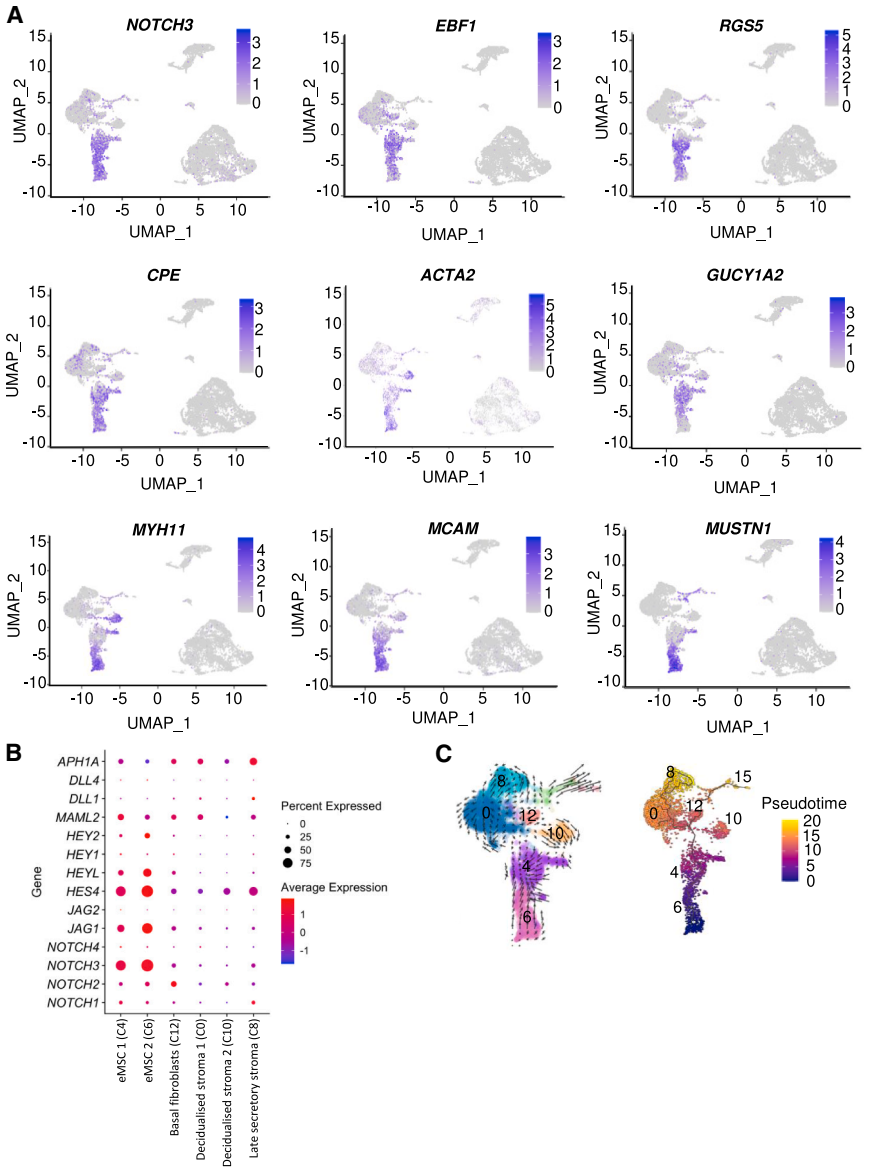


Figure 3. Two eMSC subpopulations in 6 mesenchymal cell populations identified in human endometrial stem/progenitor cell-enriched endometrial cells

(A) UMAP plots showing genes highly expressed in the 2 eMSC clusters (C4 and C6).

(B) Dot plot showing *NOTCH* signaling genes in the endometrial mesenchymal populations.

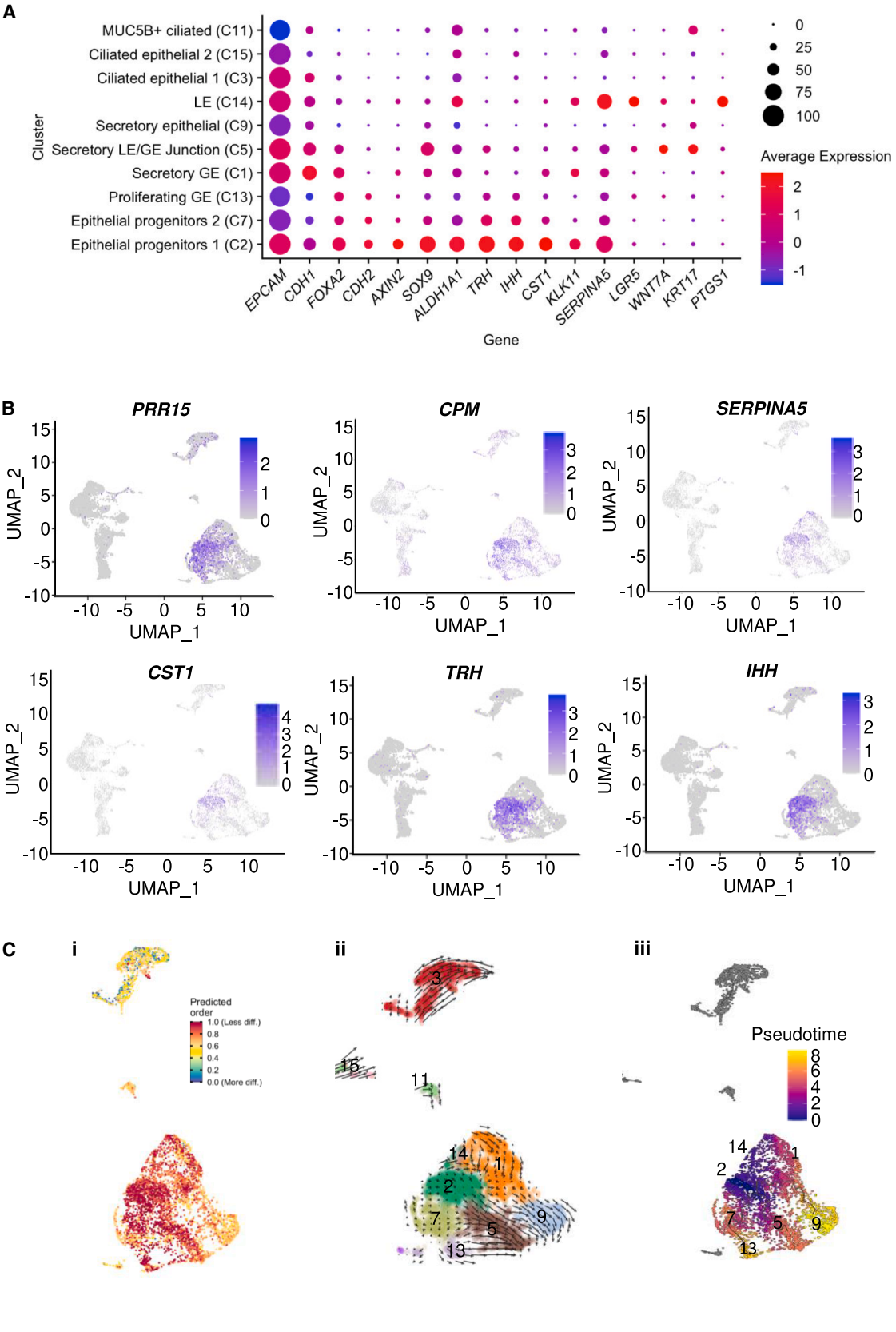
(C) Cell fate trajectory of mesenchymal populations; left: RNA velocity based on spliced and non-spliced transcripts using velocyto and scVelo. Velocity field projected onto UMAP overlayed with arrows showing the local average velocity evaluated on a regular grid; right: the predicted cell fate trajectory using Monocle3, setting C6 as the root node containing the largest proportion of *SUSD2*$^{+}$*MCAM*$^{+}$ cells.

UMAP, uniform manifold approximation and projection.

See also Figures S3–S5 and Tables S2–S6.

(Figure 4B). Basalis epithelial progenitor marker genes *CDH2*, *SOX9*, *ALDH1A1*, and *AXIN2* (Salamonsen et al., 2021) were concentrated in C2 and extended into C7 (Figures 1E, 2, 4A, and S3E), as were steroid hormone receptors *PGR* and *ESR1* and their co-regulators (Figure S6A). Genes with higher specificity to C2 than C7 included the serine protease inhibitor *SERPINA5*, protease inhibitor *CST1*, thyrotropin-releasing hormone *TRH*, and morphogen Indian Hedgehog *IHH*. *TRH*, *CST1*, and *IHH* were the most highly expressed genes in C2 (Data S1). These epithelial progenitor signature genes except for *SERPINA5*, *CST1*, and co-regulators *NCOA2* and *NCOR1* were reported previously (Marečková et al., 2024).

CytoTRACE predicted, based on the numbers of expressed genes/cell as a determinant of developmental potential, that C2 epithelial cells had the least differentiated state and transitioned to more mature epithelial populations (Figure 4Ci). RNA velocity showed C2 and C7 as epithelial progenitor populations with more immature cell types (Figure 4Cii), specifically *CDH2*$^{+}$ and *SOX9*$^{+}$ cells (Figures S5B and S5C). Cell fate trajectory analysis setting C2 and C7 as root populations predicted differentiation toward more mature GE and LE in C1, C5, C9, and C13 cells (Figure 4Ciii). Most C2 epithelial progenitors were in G1 phase, with a few in S phase (Figure S5D). C7 epithelial progenitors had more S phase cells undergoing DNA synthesis, suggesting a transit amplifying role and more



(legend on next page)

differentiated state than C2 cells in the epithelial progenitor cell hierarchy. These data, together with gene expression analysis, suggest C2 as a basalis epithelial progenitor population. This was validated by TRH immunolocalization in basalis GE and absence in the functionalis (Figure 5A) and similarly for IHH (Figure 6B). Members of the thyroid signaling pathway up- and downstream of TRH showed a high expression of *DIO2* in C2, C12, C0, and C8, of *THRA* in C6 eMSC-2, C12 basal fibroblasts, and C10, and of *NCOA1* in all those cell types (Figure 5B). This suggests that TRH secreted from epithelial progenitors may stimulate thyroid signaling in other cell types of the basalis endometrium. Biological processes enriched for C2 genes included negative regulation of apoptotic signaling and gland development and for C7 genes were signal transduction, gland development, and regulation of cell proliferation (Figure S4C).

Endometrial epithelial stem/progenitor cell interactions in the basalis niche

To define the endometrial epithelial progenitor cell niche, LIANA (ligand-receptor analysis framework) was used to predict cell-cell interactions between cells in C2 and other clusters, revealing the potential role of the WNT signaling pathway. Signaling interactions were predicted between C8 late secretory stroma expressing *WNT5A* and *SFRP1* and basalis C2 epithelial progenitor-1 expressing *FZD5*, *3*, *1*, and *6* receptors (Figure 6Ai). Similarly, WNT signaling may be potentiated through interactions between C12 basalis niche cells expressing *RSPO-3* and C2 epithelial progenitor-1 expressing *RNF43, ZNRF3*, and *LGR4*, as well as between ligand *THBS2*, also in perivascular C4 and C6 eMSCs, and CD36-expressing C2 cells. Other interactions were predicted between *IHH* ligand from C2 epithelial progenitor-1 and receptors *PTCH1*/smoothened *(SMO)* on C12 basal stromal fibroblasts and C8 late secretory stroma, and between *SCGB3A1* and *NOTCH3* receptors on C4 and C6 eMSCs (Figure 6Aii). Similarly, *IHH* expressed in C2 epithelial progenitors-1 interacted with Hedgehog (HH) co-receptor *BOC* expressed in C12 basalis stromal fibroblasts and C8 late secretory stroma (Figure 6Aii). IHH immunolocalized to endometrial basalis epithelium in the proliferative phase, functionalis epithelium in the secretory phase, and to some stromal cells and myometrium in both phases (Figure 6B). The epithelial progenitor markers CDH2 (N-cadherin) and SSEA1 co-localized with IHH in basalis GE (Figure 6C), and BOC and SSEA1 co-localized in the basalis GE and LE, suggesting that BOC has a potential role in basalis epithelial cell migration signaling (Kim et al., 2020) to regenerate a new LE each menstrual cycle (Figure 6D).

Identification of HH signaling in endometrial cell populations showed that *IHH* is the main HH member in the endometrium and mostly expressed in epithelial progenitors (Figure 6E). *SHH* weakly is expressed in the secretory epithelial cells (Figure 6E). As well as *BOC*, *CDON* and *PTCH1*, other genes involved in HH signaling found in endometrial cell clusters include *JAG1*, *VEGFA*, *BCL2*, *CCND2*, *GAS1*, and *GLI-1* and *-2*, mostly in stromal clusters, suggesting potential signaling interactions between epithelial progenitors and basal fibroblasts and eMSCs (Figure 6E).

DISCUSSION

This study showed that the human endometrium contains at least four major stem/progenitor cell states of epithelial and mesenchymal origins that transition to mature cell states to regenerate and repopulate the endometrial functionalis following menstruation. Most previous scRNA-seq analyses of the human endometrium have not identified the epithelial progenitor cell populations as they did not involve sampling of the basalis and/or enriched cell populations using known stem/progenitor markers (Fonseca et al., 2023; Garcia-Alonso et al., 2021; Lv et al., 2022; Tan et al., 2022; Wang et al., 2020). Another scRNA-seq analysis of three adenomyosis hysterectomy samples found a progenitor cluster of ~200 cells from one patient, although they may not have been epithelial cells, as no details on harvesting the basalis glands were included (Yildiz et al., 2023). Here, we show the efficacy of our novel approach for characterizing rare human endometrial stem/progenitor populations and their niches

Figure 4. Two epithelial progenitor cell types in 10 epithelial populations identified in human endometrial stem/progenitor cell-enriched endometrial cells

(A) Dot plot showing the expression of endometrial epithelial and epithelial progenitor markers in the epithelial cell populations. C2 (epithelial progenitor-1) has the highest expression of *CDH2*, *SOX9*, *AXIN2*, and *FOXA2*.

(B) UMAP plots showing highly expressed genes in epithelial progenitor-1 (C2) and -2 (C7).

(C) Cell fate trajectory of epithelial cell clusters. (i) UMAP showing the distribution of CytoTRACE scores among epithelial cells. Green-blue indicates low stemness, more differentiated; red indicates high stemness, less differentiated; (ii) RNA velocity based on spliced and non-spliced transcripts using velocyto and scVelo. Velocity field projected onto UMAP overlayed with arrows showing the local average velocity evaluated on a grid; (iii) predicted cell fate trajectory using Monocle3, setting C2 as the root node containing the largest proportion of $CDH2^{+}SOX9^{+}$ cells.

See also Figures S3–S6 and Tables S3–S6.

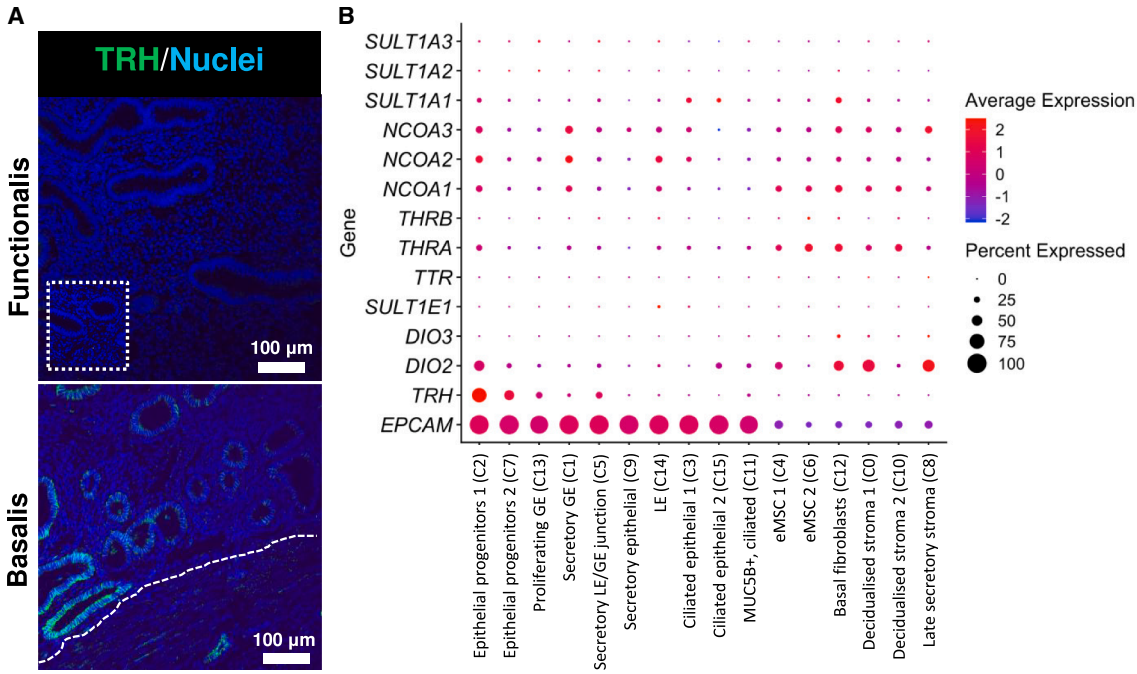


Figure 5. TRH protein and gene expression in human endometrium

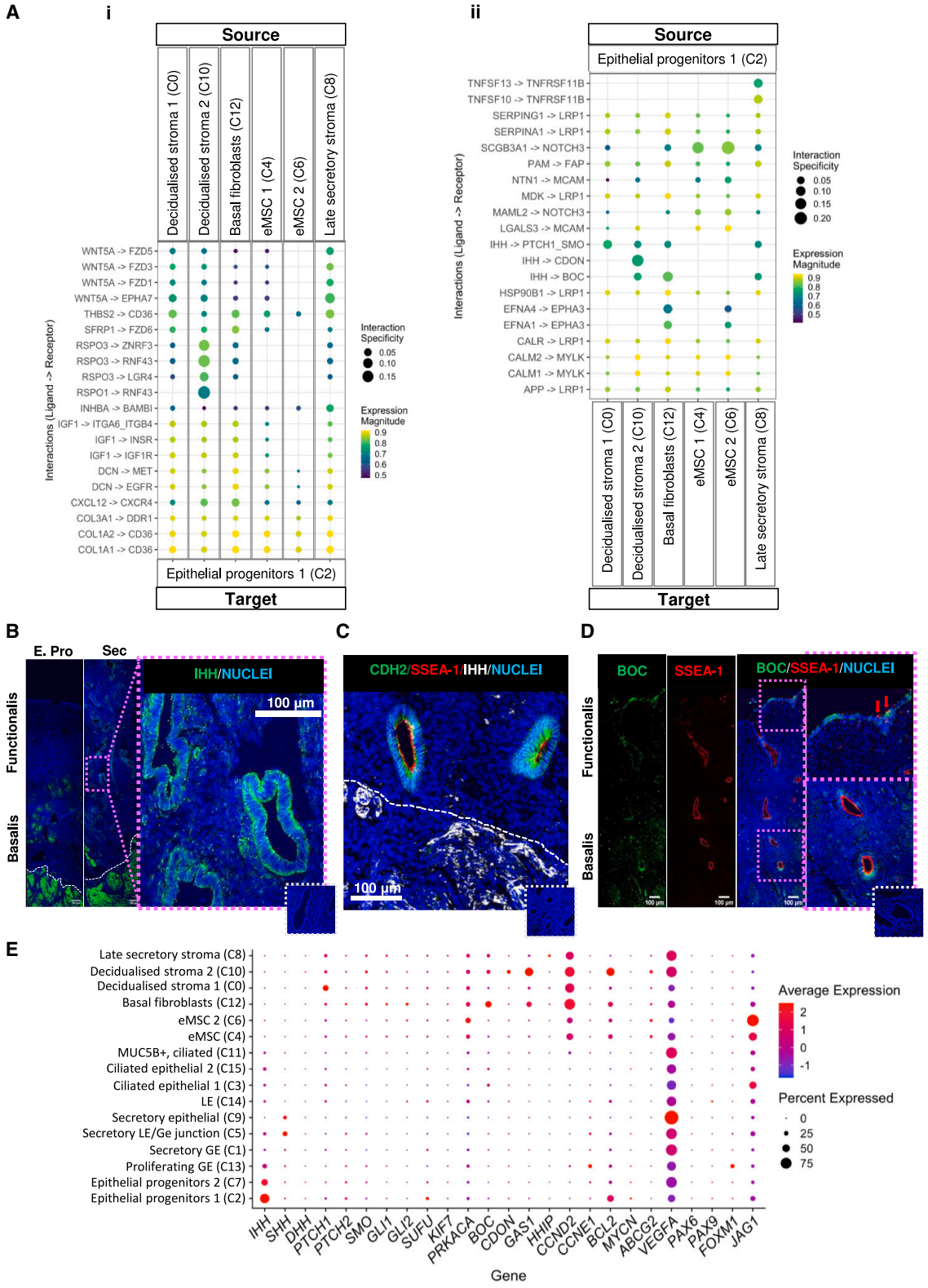
(A) Immunolocalization of TRH to basalis glands and absence in the functionalis in hysterectomy endometrium. Representative image of $n = 5$ proliferative and $n = 3$ secretory phase endometrial samples. White dashed line, endometrial-myometrial border; white dotted line, isotype control inset. Scale bars, 100 μm.

(B) Dot plot showing genes involved in thyroid signaling and their expression in the 16 endometrial cell clusters.

by FACS enrichment, using published markers of functional endometrial stem/progenitor cells and performing scRNA-seq analysis. We identified 796 *EPCAM*$^+$*CDH2*$^+$ epithelial progenitors separating into 2 clusters from 11,731 cells from five patient samples. Of these, 476 also expressed *SOX9* (*EPCAM*$^+$*CDH2*$^+$*SOX9*$^+$), mainly in the same 2 clusters. In comparison, the Human Endometrial Cell Atlas (HECA) (Marečková et al., 2024) comprising >313,000 cells identified 355 *SOX9*$^+$(*CDH2*$^+$) basalis epithelial cells from 8 of 63 patients, only 3 cells from 3 hysterectomies, and 342 cells from a single biopsy. Some of HECA's *SOX9*$^+$*(CDH2*$^+$*)* cells may be EPCAM$^-$ and of stromal origin. We also identified 12 other populations of endometrial cell types.

We confirmed two major populations of ciliated epithelial cells distinct from other cell types (Garcia-Alonso et al., 2021; Wang et al., 2020). Our two cell types had a low expression of the GE marker *FOXA2*. Four distinct GE populations were assigned based on *FOXA2* expression: one proliferating with high *MKI67* expression, one secretory, and two epithelial progenitor cell types. For the first time, we identified some secretory cells indicative of a junctional zone located between the GE and LE at gland openings with high expressions of *FOXA2*, the LE markers *KRT17* and *WNT7A*, and the secretory markers *CXCL8*, *PAEP*, and *LIF*. One secretory epithelial population with high expressions of secretory phase genes, *LIF*, *PAEP* and *SPP1*, and also the LE markers, *WNT7A* and *LGR5*, suggests this LE subpopulations may be luminal progenitors due to their *SOX9*$^+$ expression (Garcia-Alonso et al., 2021; Tan et al., 2022). The LE is likely derived from SOX9$^+$ basalis epithelial cells (Salamonsen et al., 2021).

Our study found two distinct perivascular eMSC subpopulations expressing classic MSC markers, including SUSD2; both subpopulations expressed *NOTCH3* and signaling pathway members, as reported in cultured human SUSD2$^+$ eMSCs undergoing decidualization differentiation (Murakami et al., 2014). Another hysterectomy study found two eMSC states; however, SUSD2$^+$ cells were not reported (Yildiz et al., 2023). *MUSTN1* was almost exclusively enriched for eMSC-2 (C6), and *GUCY1A2* for eMSC-1 (C4), signature markers not previously described for eMSCs. eMSC-2 are likely more immature, perivascular cells expressing similar genes (*MYH11*, *SOD2*, and *SLIT3*) as the pericyte-2 population (Queckbörner et al., 2021) and myometrial SMC (mSMC) (*ACTA2*) and myometrial PeriVascular (mPV) clusters of the HECA (Marečková et al., 2024). We suggest that the mSMC and mPV cells in the HECA are more likely eMSCs than myometrial cells as



(legend on next page)

the myometrium is not sampled in biopsies and poorly dissociates in endometrial processing protocols for hysterectomies. eMSC-1 (C4) had a closer profile to pericyte-1 (Queckbörner et al., 2021) and both ePV1b and ePV2 of the HECA. Despite varying terminologies used, considerable consensus on eMSC gene profiles exists between ours and published studies.

We found two major endometrial epithelial progenitor cell states, expressing key genes previously identified as protein markers of endometrial epithelial progenitor cells: *CDH2*, *AXIN2*, and *SOX9*. Our methods allowed us to fully characterize both states to generate a human endometrial epithelial progenitor molecular signature. It comprises the most *CDH2*$^+$ epithelial cells identified to date in endometrial scRNA-seq studies and most hysterectomy samples (n = 5) including the basalis where these cells reside. It indicates the strength of our methodology and highlights the difficulty in capturing epithelial progenitor cells from biopsies. Both endometrial epithelial progenitor cells showed high *ESR1* and *PGR* expressions correlating with protein localization in basalis endometrium across the menstrual cycle. In contrast, functionalis ESR1 and PGR localization in GE and LE varied across the cycle in response to circulating estrogen and progesterone (Gargett et al., 2008). Functionalis epithelium sequentially proliferates and differentiates in response to these hormones, but no such response occurs in the basalis. Rather, non-proliferating SSEA1$^+$SOX9$^+$ epithelial cells rapidly migrate from protruding gland stumps on days 2–3 of menstruation (Salamonsen et al., 2021). Once resurfaced, epithelial cells in the gland necks proliferate independently of estrogen on days 3–4 of menses but not in the N-cadherin$^+$ deep basal region of those same glands despite high ESR1 expression (Ferenczy et al., 1979). We identified expression of the nuclear coactivators (*NCOA2, 7*) and co-repressors (*NCOR1, 2*) that influence ESR1 transcription in the basalis epithelial progenitor clusters. The low proliferative response in SSEA1$^+$SOX9$^+$ basalis epithelial and stromal cells may be due to co-repressor activity. The role of co-activator/repressor interactions in the dichotomous response between functionalis and basalis endometrial cell ESR1 is currently unknown.

Little is known about *TRH* expression in the endometrium, but its high specificity in basalis epithelial progenitors suggests a role for thyroid hormone signaling in endometrial development and monthly tissue regeneration. TRH acts on anterior pituitary TRH receptor (TRHR) to release thyroid-stimulating hormone (TSH), which stimulates the thyroid to release T3 and T4 (Brown et al., 2023), but its role in the endometrium is unknown. TRH has been reported in the proliferative phase and in postmenopausal endometrium (Zieba et al., 2015). The macaque uterus, which also menstruates, expresses all receptors involved in thyroid hormone function, suggesting that the human uterus is likely similar with the local production and regulation of thyroid signaling (Hulchiy et al., 2012). Patients with endometriosis and infertility have increased serum prolactin (PRL) levels following TRH treatment (Cunha-Filho et al., 2002). Infertile women with hypothyroidism have higher PRL levels associated with abnormal menstrual cycle patterns (Fröhlich and Wahl, 2019). The expression of *TRH* and its receptors in the same tissue suggests local paracrine action. Although *TRHR* was not detected in our data or the HECA, we found the expression of *DIO2*, which converts T4 to active T3 (Ambrosio et al., 2017), in epithelial progenitors, basal fibroblasts, and late secretory stromal cells, suggesting that T3 maintains endometrial homeostasis. DIO2 is highly expressed in differentiating myoblasts and has a role in differentiating activated satellite (stem) cells.

Our two endometrial epithelial progenitor populations had enriched expression of *IHH*, with independent validation at the protein level. HH signaling is a key developmental pathway, and IHH is 1 of the 3 ligands that bind to PTCH1/PTCH2 receptors near the primary cilium present on most cells, including human endometrial stromal cells. IHH alleviates PTCH inhibition of SMO, activating downstream GLI transcription factors (Walterhouse et al.,

Figure 6. Ligand-receptor interactions between human endometrial epithelial progenitor cells and mesenchymal cells

(A) Top 20 interactions between ligands based on rank aggregate from (i) mesenchymal populations and receptors found on epithelial progenitors and (ii) epithelial progenitors (C2) and receptors found on mesenchymal cell types. The expression magnitude as estimated by SingleCellExperiment's LRscore and interaction specificity was determined by NATMI's edge specificity weights from 0 to 1, where 1 means that both the ligand and receptor are uniquely expressed in each pair of cells.

(B) Immunolocalization of IHH in proliferative (Pro)- and secretory (Sec)-phase endometrium.

(C) Immunoco-localization of CDH2 (N-Cadherin), SSEA1, and IHH in basalis glands.

(D) Immunoco-localization of BOC and SSEA1 in hysterectomy endometrium in luminal epithelium (upper inset) and basalis glands (lower inset). Red arrows show SSEA1 LE immunostaining.

(E) Dot plot showing the expression of Hedgehog (HH) signaling genes in endometrial cell populations.

(B–D) show representative examples of n = 5 proliferative and n = 3 secretory phase endometrial tissues; scale bars, 100 μm; white dotted line, isotype control inset; white dashed line, endometrial-myometrial border.

2003). In mice uteri, Ihh regulates progesterone signaling between the epithelium and stroma (Lee et al., 2006). Conditional Smo knockout shortens diestrus, results in uteri thickening failure in estrogen-dependent cycle stages, abnormal endometrial morphology, and infertility (Roberson et al., 2023). We identified cell interactions between the epithelial progenitor IHH and the basalis stromal IHH co-receptor BOC, PTCH1, and the GLIs. In mice, primordial germ cells migrating from the hindgut to the genital ridges express both SSEA1 and Boc (Kim et al., 2020), suggesting a possible role of BOC in migration and re-epithelization of basalis epithelial progenitors over the raw endometrium as it repairs during menstruation (Salamonsen et al., 2021). Together, these findings suggest that IHH may regulate endometrial epithelial migration, repair, and regeneration during menstruation, possibly maintaining the stem cell niche.

Limitations that may impact the interpretation of our findings include the small sample size and variability among patient samples, differences in cycle stage, age, and pathologies, which may influence the cell composition and gene expression. The choice of clustering resolution, while computationally optimized and biologically informed, may not fully align with all biologically meaningful cell types and states. Validation in larger, more diverse datasets will confirm the robustness and broader applicability of the results. Secondly, lineage tracing was not performed to show that our epithelial progenitor populations mediate endometrial regeneration. However, recent mapping of the somatic mutation signatures of individual human endometrial glands showed their clonality by whole-genome sequencing (WGS)/whole-exome sequencing (WES) (Moore et al., 2020; Yamaguchi et al., 2022). The vertical functionalis glands arising from the horizontal segments of basalis glands, where N-cadherin^{+} epithelial progenitors reside, had the same mutation signature, indicating that these basalis progenitors regenerate functionalis glands each month. Furthermore, trajectory-based lineage tracing and ligand-receptor interaction predictions were derived from bioinformatic analyses and should be interpreted with caution in the absence of functional validation. However, these widely used approaches produced results consistent with known biology and provide valuable hypotheses for future studies.

Characterizing gene expression profiles of endometrial stem/progenitor cell types and states at the single-cell level and further examining their niches will elucidate their role in regeneration of the endometrium during each menstrual cycle and in the pathogenesis of endometrial proliferative disorders, such as endometriosis, adenomyosis, Asherman's syndrome, and thin endometrium, possibly facilitating the identification of new targets for developing new therapies.

RESOURCE AVAILABILITY

Lead contact

Requests for further information should be directed to and will be fulfilled or facilitated by the lead contact, Caroline Gargett (caroline.gargett@hudson.org.au).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All scRNA-seq data generated in the present study have been deposited at the NCBI Gene Expression Omnibus, with accession no. GEO: GSE301344 via https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE301344 and will be publicly available as of the date of publication. No original code and/or algorithms are reported: however, code used for the data analysis can be provided upon request. Additional information required to re-analyze the data reported in this paper is available from the lead contact on request.

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AUTHOR CONTRIBUTIONS

C.E.G. and G.W.M. conceptualized the study, and along with H.C.F., S.M., and C.E.F., designed the research; H.C.F., S.M., C.E.F., K.A.W., F.L.C., and E.M.-G. performed the research; H.C.F. and S.M. analyzed data; H.C.F., S.M., and F.L.C. visualized the data and results; H.C.F., S.M., B.M., C.E.G., and G.W.M. interpreted the results; C.E.G. was the project administrator; C.E.G., J.T., and G.W.M. contributed resources; H.C.F., S.M., and C.E.G. wrote the paper. All authors have reviewed and edited the paper.

DECLARATION OF INTERESTS

L.R. declares he is a minority shareholder in the Monash IVF Group and receives consultancy fees from Merck, Organon, and Abbott; however, there was no involvement of these interests in the present study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
PE-conjugated mouse monoclonal anti-human N-cadherin (CD325) (clone 8C11) (10 μg/mL)	BioLegend	Cat# 350805; RRID: AB_10662542
BV421-conjugated mouse monoclonal anti-human/mouse SSEA-1 (CD15) (MC-480 clone) (1.25 μg/mL)	BioLegend	Cat#125614; RRID: AB_2562672
BB515-conjugated mouse monoclonal anti-human EpCAM (CD326) (EBA-1 clone) (1/80 dilution)	BD Biosciences	Cat#565398; RRID: AB_2728107
APC-conjugated mouse monoclonal anti-human SUSD2 (W5C5 clone) (1.25 μg/mL)	BioLegend	Cat#327408; RRID: AB_256188
PECy7-conjugated mouse monoclonal anti-human PECAM (CD31) (WM59 clone) (1/80 dilution)	BD Biosciences	Cat#563651; RRID: AB_2738348
PECy7-conjugated mouse monoclonal anti-human CD45 (HI30 clone) (1/80 dilution)	BD Biosciences	Cat#557748; RRID: AB_396854
Rabbit anti-human N-cadherin (D4R1H clone)	Cell Signaling Technology	Cat#13116; RRID: AB_2687616
Mouse anti-human SSEA-1 monoclonal antibody (MC-480 clone) (1/100)	Merck (Chemicon)	Cat#MAB4301; RRID: N/A
Rabbit anti-human TRH (1/500)	Sigma-Aldrich	Cat#HPA035596; RRID: N/A
Rabbit anti-human IHH (1/100)	Abcam	Cat#Ab52919; RRID: AB_2121994
Rabbit anti-human BOC (1/100)	ThermoFisher	Cat# BS-12322R; RRID: N/A
Donkey anti-rabbit Ig AF488 (1/500)	ThermoFisher	Cat#A-21206; RRID: AB_2535792
Goat anti-mouse IgG AF568 (1/500)	ThermoFisher	Cat#A-11004; RRID: AB_2534072
Donkey anti-rabbit IgG AF647 (1/500)	Abcam	Cat#Ab150075; RRID: AB_2752244
Biological samples		
Fresh human hysterectomy tissue	In house Ritchie Center human tissue bank	N/A
PFA-fixed OCT frozen hysterectomy tissue block	In house Ritchie Center human tissue bank	N/A
Chemicals, peptides, and recombinant proteins		
Collagenase Type I	Worthington Biochemical	EC: 3.4.24.3; CAS: 9001-12-1
DNAse I	Worthington Biochemical	EC: 3.1.21.1; CAS: 9003-98-9
Propidium Iodide	BD Biosciences	556463; CAS:25535-16-4
Hoechst 33258 (5 μg/mL)	ThermoFisher	H1398; CAS: 23491-45-4

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
Compensation Plus Particle Set Beads	BD Biosciences	Cat#560497
Chromium Single-cell 3′ GEM, Library & Gel Bead Kit v3	10× Genomics	Cat#PN-1000075
Deposited data		
Raw and analyzed single cell RNA-seq data	This paper	GEO: GSE301344
Software and algorithms		
FACS Sortware Software v1	BD Biosciences	N/A
Cell Ranger v6.0.1	(Zheng et al., 2017)	https://www.10xgenomics.com/support/software/cell-ranger/latest
R v4.1.2	(R Development Core Team, 2010)	https://www.r-project.org/
DoubletFinder	(McGinnis et al., 2019)	https://github.com/chris-mcginnis-ucsf/DoubletFinder
Seurat v4.1.1	(Hao et al., 2021)	https://satijalab.org/seurat/
scTransform	(Hafemeister and Satija, 2019)	https://github.com/satijalab/sctransform
Harmony	(Korsunsky et al., 2019)	https://github.com/immunogenomics/harmony
clustree v0.5.1	(Zappia and Oshlack, 2018)	https://github.com/lazappi/clustree
Monocle3	(Cao et al., 2019)	https://cole-trapnell-lab.github.io/monocle3/
velocyto	(La Manno et al., 2018)	https://velocyto.org/
scVelo	(Bergen et al., 2020)	https://scvelo.readthedocs.io/en/stable/
CytoTRACE R package v0.3.3	(Gulati et al., 2020)	https://cytotrace.stanford.edu/
LIANA	(Dimitrov et al., 2022)	https://github.com/saezlab/liana
Other		
ShinyGO v0.76	Ge et al. (2020)	https://bioinformatics.sdstate.edu/go/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Sample collection

Endometrial tissues containing the functionalis and basalis were collected from women <50 years undergoing hysterectomy (n = 5) for benign uterine conditions (Table 1). Exclusion criteria were previous endometrial ablations, thickened endometrium, endometrial cancer and taking hormone therapy. Eligible women gave written informed consent. Human ethics approval was obtained from Monash Health HREC and the US Department of Defense OHRO (RES-19-0000-388A/E00700.1a.1b). Endometrial samples were fixed in 4% PFA overnight at 4C (n = 8) or placed into cold DMEM:F12 collection medium containing 5% newborn calf serum and antibiotics/antimycotic and immediately transported at 4C to the laboratory for processing (n = 5).

METHOD DETAILS

Isolation of primary endometrial cells

Endometrial tissue was digested to single cells within 18 h. The functionalis was removed by scraping the tissue off the myometrium and finely minced, then the full basalis recovered by dissecting 1–2 mm myometrium with attached

endometrium (visible microscopically) and finely dicing the tissue (Nguyen et al., 2017) (Figure 1A). Minced tissue underwent a 2-step enzymatic/mechanical dissociation with vigorous trituration every 5 min at 37C in PBS (Gibco) containing collagenase type I (5 mg/mL; Worthington), DNAse I (0.08 mg/mL; Worthington) and glucose (5mM; Merck), then strained (40 μM sieve, Falcon) to separate stromal cells from epithelial gland fragments. Red blood cells in the stromal fraction were removed by Ficoll-paque (Sigma-Aldrich) density gradient. Epithelial cells were further dissociated with collagenase type II (0.8 mg/mL, Worthington) and DNAse I (0.16 mg/mL), then strained using a 40 μM sieve. The single cell fractions were counted and recombined for FACS staining.

Endometrial stem/progenitor cell enrichment by FACS

Single cell suspensions were blocked with Fc Receptor blocker (FcR; 2 μL/10^6 cells, Miltenyi Biotec) and Rat IgG (4.4 μg/10^6 cells; Jackson ImmunoResearch). Cells were distributed in tubes as unstained, viability and Fluorescence minus 1 (FMO) controls (5×10^5 cells in 20 μL) and the full antibody panel (remaining cells at 5×10^5/20 μL at the same final antibody concentrations) and incubated with conjugated antibodies, PE-CD325 (N-cadherin) (10 μg/mL), BV421-CD15 (SSEA1) (1.25 μg/mL; BioLegend), APC-SUSD2 (1.25 μg/mL; BioLegend), BB515-CD326 (EpCAM) (0.25 μL/5×10^5 cells, BD Biosciences), PECy7-CD31 (PECAM1, endothelial exclusion marker) (0.25 μL/5×10^5 cells, BD Biosciences) and CD45 (leukocyte exclusion marker) PECy7-CD45 (0.25 μL/5×10^5cells, BD Biosciences), for 25 min at 4°C. Propidium iodide (PI; 62.5 ng/mL; BD Biosciences) was added 10 min before sorting (Wyatt et al., 2021). Compensation Plus Particle Set beads (60 μL/test; BD Biosciences) were used as single staining protocols with the above antibodies following manufacturer's instructions. Labeled endometrial cells underwent FACSorting of endometrial stem/progenitor cell subpopulations and their differentiated progeny (epithelial (EpCAM+) progenitors N-cadherin$^+$/SSEA1$^-$; N-cadherin$^+$/SSEA1$^+$; N-cadherin$^-$/SSEA1$^+$; mature epithelial cells, EpCAM$^+$/N-cadherin$^-$/SSEA1$^-$) (Nguyen et al., 2017) and EpCAM-eMSC (SUSD2$^+$; SUSD2$^-$ fibroblasts) (Masuda et al., 2012) using an Influx Cytopeia-BD 5 laser sorter and BD FACS Sortware software (Monash University FlowCore). The gating strategy for sorting viable PI$^-$CD45$^-$ cell subpopulations is shown in Figures 1A and S1. The 6 subfractions were recombined in approximately equal cell numbers.

Single-cell RNA-sequencing

The 6 cell fractions were tagged using TotalSeq A anti-human hashtags. The stem/progenitor cell populations were pooled and mature epithelial and stromal cells added for scRNAseq for each of 5 biological samples (Figure 1A) to generate barcoded single-cell 3′ gene expression libraries with the Chromium Single-cell 3′ GEM, Library & Gel Bead Kit v3 (10× Genomics) and their matched HTO library to resolve the hashtags (BioLegend). The libraries' quality were assessed by Agilent Bioanalyzer High Sensitivity DNA chip. Libraries were sequenced to an average depth of 384,676 reads/cell on the Illumina NovaSeq6000 at UQ. Sequence quality checks revealed that HTO library construction had failed; and were excluded from subsequent analyses. Cell type identification was instead performed using the bioinformatic approaches described below.

Processing scRNAseq data

Raw sequencing data was processed to fastq files using the mkfastq function in Cell Ranger (v6.0.1). Reads were aligned to the hg38 human reference, filtered for valid cell barcodes and unique molecular identifiers (UMIs) and counted using the count function. Doublets were identified in individual samples using the doubletFinder package in R (v4.1.2) (McGinnis et al., 2019) and removed. Single cells were then analyzed using Seurat (v4.1.1) (Hao et al., 2021) to remove cells with <200 expressed genes, >25% mitochondrial genes and genes expressed in <3 cells. scTransform (Hafemeister and Satija, 2019) normalization (Choudhary and Satija, 2022) and Harmony (Korsunsky et al., 2019) were used to correct between sample variation due to technical and biological differences.

Normalised and batch corrected data were used as input for linear dimensional reduction and Principal Component Analysis (PCA). Cells were clustered using the Louvain algorithm and the first 20 principal components in Seurat. Clustering was performed across multiple resolutions between 0.01 and 1.0 and cluster stability assessed using clustree (v0.5.1).

Cell-type annotation

Cells were annotated according to cell type using two supervised transcriptome-based cell-type classification methods. Previously published marker genes for endometrial cell types (Endometrial Epithelial Cells, EPCAM, CDH1; epithelial progenitors, SOX9, AXIN2, CDH2; glandular epithelial, FOXA2; ciliated epithelial, FOXJ1, PIFO, TPPP3; luminal epithelial, LGR5, PTGS1, WNT7A; MUC5B+ epithelial, MUC5B, TFF3, SAA1; Endometrial Stromal Cells, WT1, IGF1, PDGFRA; Perivascular, SUSD2, RGS5, NOTCH3 (Cousins et al., 2021; Wang et al., 2020) were used to assign clusters to cell types. Similarly, additional fine-scale subpopulations of stromal and SOX9+ epithelial cells were annotated using marker genes published

by Garcia-Alonso et al. including basal fibroblasts (C7, OGN, ACTA2), non-decidualised proliferative stroma (MMP11, CRABP2, ECM1), decidualised secretory stroma (IL15, CFD), non-cycling SOX9+LGR5+ LE cells (SOX9, LGR5, KRT17, WNT7A) and SOX9+LGR5-basal gland epithelial cells (SOX9, LGR5, IHH, EMID1). The expression profile of each cell was also used to assign cell type annotations by comparing them to the Human Cell Atlas (HCA) and the Blueprint and Encode reference datasets using SingleR (v2.8.0) (Aran et al., 2019).

Gene ontology analysis

Gene enrichment and ontology analysis was performed using ShinyGO v 0.76 on the most significantly increased genes found in at least 75% of cells within clusters 2 and 7, and clusters 4, 6, 0 and 8.

Trajectory and RNA velocity analysis

Transcriptional profiles were used to estimate the transition between cell states in the endometrium using cell fate trajectory analysis in Monocle3 (Cao et al., 2019). Root cells were set computationally as the trajectory graph node containing the highest proportion of cells from the 'earliest time point'. For cells of mesenchymal origin the node where most SUSD2$^+$ cells were expressed was selected. Similarly, the 'earliest time point' for cells of epithelial origin was selected as the node with *CDH2*$^+$*SOX9*$^+$, *CDH2*$^-$*SOX*$^+$ or *CDH2*$^+$*SOX*$^-$*EPCAM*$^+$ as root cells. RNA velocity was also used to assess cell trajectory using dynamic modeling with velocyto (La Manno et al., 2018) and scVelo (Bergen et al., 2020). Cellular Trajectory Reconstruction Analysis using gene Counts and Expression (CytoTRACE) predicted the differentiation state of the cells based on the number of detectably expressed genes/cell as a determinant of developmental potential. Raw single-cell expression counts were used as input in the 'CytoTRACE' function in the CytoTRACE R package v0.3.3 (Gulati et al., 2020) and results visualised using 'plotCytoTRACE' function and coordinates generated from UMAP.

Cell interaction analysis

Interactions between epithelial progenitors and mesenchymal cell types and cell communication analyses used LIANA (LIgand-receptor ANalysis frAmework) (Dimitrov et al., 2022). LIANA combines multiple tools and resources and provides a rank aggregate from the results of different methods serving as a consensus across methods. Analyses were run using epithelial progenitors, C2, as the target and mesenchymal clusters as the source and again with C2 as the source. The top 20 interactions based on the rank aggregate were plotted.

Immunofluorescence

Immunofluorescence was performed on 8 μm full thickness endometrium sections, previously fixed in 4% paraformaldehyde and cryopreserved in 30% sucrose and embedded in optimal cutting temperature compound (OCT). Unless stated all sections underwent the following protocol. Briefly, sections were thawed and rehydrated in PBS, and blocked in DAKO blocking solution (Agilent X0909) for 1 h at room temperature (RT). Primary antibodies for Rabbit anti-N-cadherin (Cell Signaling Technolgy 13116, 1/100), Mouse anti-SSEA-1 (Merck MAB4301 1/100), Rabbit anti-IHH (Abcam ab 52919 1/100), Rabbit anti-TRH (Sigma-Aldrich HPA035596 1/500) and Rabbit anti-BOC (ThermoFisher BS12322R 1/100) were applied in 1%BSA:PBS overnight at 4°C. Rabbit IgG (ThermoFisher 10500C) and Mouse IgM (Caltag MGM00) isotype controls were used at the same concentrations as primary antibodies. Sections were incubated with secondary antibodies donkey anti-rabbit AF488 (Thermofisher A21206 1/500), goat anti-mouse AF568 (ThermoFisher A11004 1/500), or donkey anti rabbit AF647 (Abcam ab 150075 1/500) for 2h RT. In the case of double or triple immunostaining, antibodies were added sequentially and specific species blocks (20% goat serum:1%BSA:PBS or 20% donkey serum:1%BSA:PBS) were applied prior to the additional primary antibodies for 30min RT. Nuclei were stained with 5 μg/mL Hoechst 33258 diluted in PBS for 5 min RT. Slides were mounted in fluorescent mounting medium (DAKO S3203) and imaged using an Olympus FV1200 confocal microscope using a 20× objective lens. To obtain full thickness images, multiple fields of view were stitched together using the tile and stitch functions. Brightness and contrast were adjusted in a linear manner using FIJI software.

QUANTIFICATION AND STATISTICAL ANALYSIS

Differentially expressed genes (DEGs) between clusters were identified using the 'FindMarkers' function in Seurat using Wilcoxon rank-sum test and restricting tests to genes expressed in ≥25% of cells in either of the two populations compared. Expression in each cluster was compared to all other clusters combined. Gene expression differences were considered significant if they had an absolute log-fold change of ≥0.1 and adjusted p-value <0.05.